
Learnability, Reweighted: Which Tasks the Estimator Makes Active in Verifiable- Reward RL

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Abstract

RL with verifiable rewards often treats task curricula as independent of the group estimator that converts sampled outcomes into updates. We ask which tasks that estimator makes active. Every finite-group estimator induces an activity geometry over pass rates — the expected absolute mass of the coefficient vector it can emit — and for the practical MaxRL convention that drops all-fail groups this geometry has an exact closed form, $A_N(p) = 2(\text{pass@}N - \text{pass@}1)$, the estimator’s own $\text{pass@}N$ gap. It factors exactly as $p(1-p) \cdot w_{N-1}(p)$: the canonical learnability score reweighted by MaxRL’s own objective weight at the truncation the deployed estimator targets, recovering $p(1-p)$ as the $N=2$ slice, with a peak that migrates toward harder tasks as $\log N/N$ and an all-fail rule that shifts the truncation order to $N-1$. This is an estimator-side diagnostic, not a theorem of learning progress, and we test it as a hypothesis. In a fresh 20-seed Acrobot tournament at $N=16$, the deployed- N score improves target-uniform AUC over $p(1-p)$ by $+0.0480$ (95% paired-bootstrap CI $[+0.0209, +0.0738]$) and also beats uniform sampling in this fixed eight-threshold family; the effect replicates on two further platforms ($+0.0322, +0.0307$). Two preregistered follow-ups then bound the claim. Holding the deployed estimator at $N=16$ and sweeping only the score exponent, performance rises past $N=16$ (argmax at u_{64}), so the harder-peaked shape is what helps and the deployed- N peak location is not. Dropped into robust PLR on AMaze in place of MaxMC, the pure activity priority is starved of signal at one Bernoulli per level visit and does not beat the upstream baseline, while an exact-probability Digits counter-test rejects a universal estimator-to-sampler mapping. The controlled positive and these two complementary failures support a calibrated conclusion: finite-group coefficient activity is a principled, estimator-conditioned source of curriculum hypotheses, not a universal measure of learning utility.

1 Introduction

RL with verifiable rewards (RLVR) converts automatically checkable outcomes into policy updates, and every update first pays for a group of N model rollouts. A task curriculum tries to make that budget productive by sampling tasks believed to be learnable now. The dominant curriculum score in the literature is learnability, $p(1-p)$ (Tzannetos et al., 2023; Rutherford et al., 2024): sample tasks the policy solves about half the time. We ask a question that score does not: learnable for which learner? Modern RLVR does not run REINFORCE with one rollout. It runs a group estimator — MaxRL (Tajwar et al., 2026), GRPO, RLOO — that converts N binary outcomes into coefficients, and the coefficients that estimator can emit on a task are a function of the task’s pass rate and of N .

We answer for practical MaxRL. Its expected absolute coefficient mass on a task with pass rate p is exactly

$$A_N(p) = 2\{1 - (1-p)^N - p\} = 2(\text{pass@}N - \text{pass@}1),$$

and the resulting score factors as

$$u_N(p) = A_N(p)/2 = p(1-p) \cdot w_{N-1}(p),$$

where $w_T(p) = \sum_{k=1}^T (1-p)^{k-1}$ is MaxRL’s own weight function (Cor. 1). Coefficient activity is learnability reweighted by the deployed estimator. $p(1-p)$ is the $N=2$ slice; the peak $p_N^* = 1 - N^{-1/(N-1)} \simeq \log N/N$ is derived, not tuned, and moves toward harder tasks as the rollout budget grows. MaxRL asked which objective more compute should buy; we ask which tasks that objective should be fed, and the answer is written in the same weight function.

We then test the derived score as a curriculum hypothesis and report where it holds and where it stops:

1. The shape beats the learnability slice, and replicates. In a fixed Acrobot pool at deployed $N=16$, u_{16} beats $p(1-p)$ by $+0.048$ under a frozen rule, replicated on two further platforms ($+0.032$, $+0.031$); an exact-probability Digits study shows the same shape effect at $+0.208$ under MaxRL and $+0.177$ under RLOO (Fig. 2).
2. The shape helps; the peak location does not. Holding the estimator at $N=16$ and sweeping the score exponent, performance rises past $N=16$ (§4.2). The value of u_N over $p(1-p)$ is in its tail — precisely where w_{N-1} differs from 1.
3. As a signal of its own it is starved. Dropped into robust PLR on AMaze in place of MaxMC, activity does not beat upstream: one Bernoulli per level visit cannot compete with a critic read at every timestep (§4.3).
4. The estimator belongs in the curriculum’s evaluation. Digits rejects a universal estimator-to-sampler mapping; the maze factorial shows the estimator ordering coverage under either sampler; a Countdown aggregate shows a logged proxy moving opposite to raw outcomes.

The identity is an estimator-side diagnostic, not a theorem of learning progress. The paper is deliberately narrower than the project record: exploratory mechanisms, failed hypotheses, robotics pilots, gate studies, and implementation detail remain in the appendix and artifact.

2 Coefficient activity under practical MaxRL

Setup. For task x , let $r_i \in \{0,1\}$ be N conditionally independent rollout rewards with pass rate $p = p_\theta(x)$ and $K = \sum_i r_i$. Let $s_i = \nabla_\theta \log \pi_\theta(y_i | x)$ be the response score and $\hat{g} = \sum_i w_i s_i$ the group update. Then $\text{pass@}N = 1 - (1-p)^N$. We analyze the zero-stabilizer idealization of the released practical centered MaxRL convention, which assigns

$$w_i = \frac{r_i}{K} - \frac{1}{N} \quad \text{when } K > 0, \quad w_i = 0 \quad \text{when } K = 0.$$

All-pass groups also have zero contrast. The released implementation uses a finite denominator stabilizer ε ; for $K > 0$ this multiplies the ideal coefficients by $K/(K + N\varepsilon)$. Thus the identities below are exact for the stated zero- ε convention rather than for its finite- ε magnitude.

Definition 1 (Finite-group activity geometry). A symmetric binary group estimator $\mathcal{E}$ assigns coefficients $\mathbf{c}_\mathcal{E}(K) \in \mathbb{R}^N$ to a group with K successes. Its activity geometry is the expected absolute coefficient mass it emits on a task of pass rate p ,

$$\mathcal{A}_{\mathcal{E},N}(p) = \mathbb{E}_{K \sim \text{Bin}(N,p)} \|\mathbf{c}_\mathcal{E}(K)\|_1 = \sum_{k=0}^N \binom{N}{k} p^k (1-p)^{N-k} a_{\mathcal{E},k}, \quad a_{\mathcal{E},k} = \|\mathbf{c}_\mathcal{E}(k)\|_1,$$

a Bernstein polynomial in p whose coefficients are the estimator’s per-outcome masses.

Choosing an estimator fixes $\{a_{\mathcal{E},k}\}$ and therefore fixes which tasks can emit an update at all. This is orthogonal to the objective-side question of which finite-rollout objective to deploy (Tajwar et al., 2026; Zheng, 2026): we hold the deployed estimator fixed and ask which tasks its realized coefficients make active. What singles out practical MaxRL is that its Bernstein sum collapses to a closed form.

Lemma 1 (Practical truncation order). The expectation of the drop- $K = 0$ practical estimator equals the gradient of the $(N - 1)$ -term truncated maximum-likelihood objective, not the N -term objective.

For $J_T(p) = -\sum_{t=1}^T (1-p)^t/t$, direct conditioning gives

$$\mathbb{E}\hat{g} = \left(\frac{1 - (1-p)^N}{p} - (1-p)^{N-1} \right) \nabla p = \left(\sum_{j=0}^{N-2} (1-p)^j \right) \nabla p = \nabla J_{N-1}.$$

The correction matters whenever a theoretical truncation parameter is mapped to a deployed group size. This $N - 1$ statement does not revise MaxRL’s theorem for its uncentered conditional estimator; it characterizes the released practical centered convention after its all-fail group is zeroed, so the shift from N to $N - 1$ is a property of the deployed centered-and-dropped rule rather than of the maximum-likelihood expansion. Details and the non-dropped control-variate convention are in App. A.

Proposition 1 (Exact coefficient mass). The expected absolute coefficient mass of the practical estimator is

$$A_N(p) = \mathbb{E} \left[\sum_{i=1}^N |w_i| \right] = 2\{1 - p - (1-p)^N\} = 2(\text{pass@}N - \text{pass@}1).$$

For a mixed group with $0 < K < N$, successes and failures each contribute $(N - K)/N$ absolute mass. Therefore

$$A_N(p) = \frac{2}{N} \mathbb{E}[(N - K)\mathbf{1}\{K > 0\}] = 2\{(1-p) - (1-p)^N\}.$$

This identity concerns coefficients, not gradient norm, signal-to-noise ratio, or expected improvement: score-gradient geometry can change its downstream effect.

The bridge to the expected update is nevertheless exact. With $\mu_+ = \mathbb{E}[s_i \mid r_i = 1, x]$ and $\mu_- = \mathbb{E}[s_i \mid r_i = 0, x]$,

$$\mathbb{E}[\hat{g} \mid x] = u_N(p)(\mu_+ - \mu_-), \quad u_N(p) = A_N(p)/2.$$

Thus u_N is the estimator-controlled scalar factor in the expected update; the conditional score geometry remains outside the mass calculation. It also has a direct event interpretation. For any designated rollout,

$$u_N(p) = \Pr(K > 0) - \Pr(r_1 = 1) = \Pr(r_1 = 0, K > 0) = (1-p)\{1 - (1-p)^{N-1}\}.$$

Thus activity is precisely the probability that one rollout fails while at least one peer in the purchased group succeeds. This makes the two dead zones transparent: an all-fail task supplies no positive contrast, while an all-success task supplies no negative contrast. It also clarifies why N is part of the curriculum definition rather than an implementation detail.

Corollary 1 (Activity is reweighted learnability). Write MaxRL’s population gradient as $\nabla J_T = \mathbb{E}_x[w_T(p)\nabla p]$ with weight function $w_T(p) = \sum_{k=1}^T (1-p)^{k-1}$ (Tajwar et al., 2026). Then, exactly,

$$u_N(p) = p(1-p) \cdot w_{N-1}(p).$$

By Lemma 1 the deployed estimator targets $T = N - 1$, and $p(1-p) \sum_{k=1}^{N-1} (1-p)^{k-1} = p\{1 - (1-p)^{N-1}\} \cdot (1-p)/p \cdot p = (1-p)\{1 - (1-p)^{N-1}\} = u_N(p)$. So coefficient activity is the learnability score $p(1-p)$ — the variance of a single Bernoulli rollout, and the $N=2$ or pure-REINFORCE case — reweighted by MaxRL’s own weight function at the truncation order the deployed estimator actually targets. This is the sense in which the score is derived: $p(1-p)$ peaks at $\frac{1}{2}$ regardless of estimator, w_{N-1} is decreasing in p and grows with N as $p \rightarrow 0$, and their product’s peak $p_N^* = 1 - N^{-1/(N-1)}$ is where an informative rollout meets an estimator that still cares. The canonical binary-success learnability score $p(1-p)$ — used explicitly by SFL and LILO (Rutherford et al., 2024; Foster et al., 2025), and recovered here as RLOO’s coefficient mass — is therefore the $N=2$ slice of practical MaxRL activity: it scores tasks as if the learner were REINFORCE with one rollout, while u_N scores them for the estimator that is deployed.

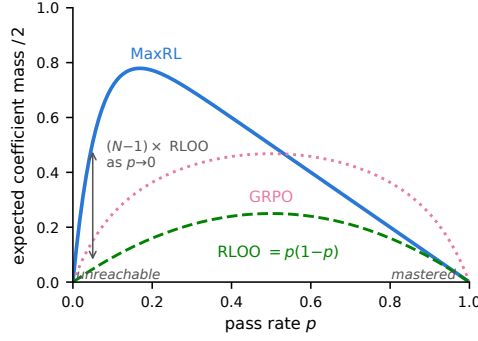


Figure 1: Three estimators, three activity geometries. Exact expected half-mass at $N = 16$ under zero-stabilizer MaxRL, gradient-averaged RLOO, and the zero-stabilizer sample-SD-normalized GRPO convention. Coefficient mass is sign-blind, so the curves motivate rather than prove the empirical coverage ordering.

Across estimators. As $p \rightarrow 0$, MaxRL carries $(N - 1)$ times the coefficient mass of standard $1/N$ -averaged RLOO; as $p \rightarrow 1$ both decay at first order in $1 - p$, whereas sample-SD-normalized GRPO allocates relatively more mass to near-mastered prompts (App. A).

Fixed-completion check. We tested the arbitrary- N prediction on a 36-task CPU skill-chain testbed with $N \in \{2, 4, 8, 16, 32\}$, eight paired seeds, no recycling, and exactly 51,200 sampled completions per cell; teacher decay was normalized per completion so changing N did not silently change its memory horizon. At that fixed budget the mean normalized pass-rate AUC difference $u_N - p(1 - p)$ is .0000, +.0307, +.0920, +.1526, +.1909 across those N , and u_N - uniform is -.0106, -.0032, +.0309, +.0453, +.0836. The $N = 2$ identity reproduces bit-for-bit, and for every $N > 2$ the rollout-aware utility beats the score-only $p(1 - p)$ comparator in all eight paired seeds. The mean gap grows across tested N , but only 5/8 seed trajectories are monotone, so we claim no scaling law; nor does u_N universally beat uniform, whose contrast turns positive only at $N \geq 8$ (positive-seed counts 3/8, 3/8, 8/8, 7/8, 7/8). Seed signs are descriptive and the sweep was not preregistered: this is evidence for rollout awareness in one co-designed testbed, not a blanket curriculum advantage.

Exact-probability counter-test. The stronger test does not generalize the sampler mapping. Under an internally hashed local protocol without an immutable public pre-execution commit, a fresh 24-block Digits contextual bandit computes the policy’s exact correct-class probability before every update. Its internally frozen estimator-by-sampler interaction, +.01589 (95% CI [-.01686, +.04712], exact sign-flip $p = .350$; 15/24), is not supported: MaxRL favors u_8 over $p(1 - p)$ by +.20842 (23/24), but so does RLOO (+.17665, 24/24), reversing its prediction, and both estimator-matched samplers fall below uniform. One study thus carries the paper’s largest shape effect and its sharpest negative — the harder-peaked score beats $p(1 - p)$ under both estimators, while the estimator-matching rule that would make it a mapping is rejected. The activity algebra generates testable hypotheses, not a universal learnability or curriculum-optimality law.

3 From mass to interventions

Task sampling. FrontierMax operationalizes the identity without changing the within-group estimator. Each task maintains discounted Beta evidence, draws a posterior pass rate $\tilde{p}_x$, and samples approximately as

$$q(x) = \frac{\epsilon}{M} + (1 - \epsilon) \frac{[u_N(\tilde{p}_x)]^\gamma}{\sum_{j=1}^M [u_N(\tilde{p}_j)]^\gamma}.$$

The floor ϵ , concentration γ , decay, and posterior estimator remain empirical choices; the theory contribution is the utility and its N -dependence, not a parameter-free scheduler.

Table 1: Contribution–evidence map. Status is attached to the specific object evaluated, not transferred across rows.

Object	Claim/mechanism	Evidence allowed here
Coefficient identity	Exact A_N , $T = N - 1$, and posterior-integrated u_N for the stated practical-MaxRL convention	Proved; estimator activity only, not learning progress
Direct fixed-pool teacher	Thompson-scored u_N with proportional sampling and a uniform floor	Scoped empirical; Acrobot supports the score in one fixed family, while Digits rejects a universal mapping
Coefficient-Activity PLR	Analytic posterior activity replaces MaxMC priority inside robust PLR; replay and staleness remain PLR mechanisms	Development negative (§4.3); as a replacement for MaxMC it does not beat upstream robust PLR on AMaze at 5,000 updates

The same utility also admits an exact deterministic posterior priority. If $p_x \mid \mathcal{D} \sim \text{Beta}(a_x, b_x)$ and $(z)_N$ denotes the rising factorial, then

$$\mathbb{E}[u_N(p_x) \mid \mathcal{D}] = 1 - \frac{(b_x)_N}{(a_x + b_x)_N} - \frac{a_x}{a_x + b_x}.$$

This is distinct from the Thompson-sampled rule above. Because u_N is strictly concave for $N > 1$, scoring the posterior mean instead systematically overstates expected activity by a nonnegative closed-form uncertainty penalty (App. A). Which posterior treatment learns best remains an empirical question.

Contribution and evidence boundary. Coefficient activity is the shared quantity, not a license to conflate a proved identity, a tested fixed-pool teacher, and an untested replay overlay; Table 1 makes the evidence ladder explicit.

Predictions. The mass curves motivate, but do not prove, an empirical expectation: MaxRL and variance-normalized GRPO can exhibit different coverage orderings under a common sampler. This need not imply a sampler-by-estimator interaction and requires controlled measurement.

4 Evidence

The evidence has two roles: the Acrobot tournament isolates the deployed- N score shape under a fixed scaffold, while the maze factorial tests an estimator-conditioned coverage ordering on real neural gradients — an estimator-side prediction, not the score. The Digits counter-test supplies both the boundary on estimator-matching and, read the other way, the largest shape effect in the paper. The score’s positive support is deliberately small-scale and fully controlled. Figure 2 states for each contrast what it is and is not: one frozen confirmatory primary, concordant supporting measurements, and two preregistered boundaries.

4.1 Fresh fixed-pool Acrobot score test

We isolate the arbitrary- N score on official Gymnasium Acrobot-v1. Eight nested tip-height predicates share one task-blind H64 actor (640 parameters), the hardest being exactly native Acrobot success. All arms use practical MaxRL at $N = 16$, no hindsight, learning rate 3×10^{-4} , and a nominal two million paid environment transitions; twenty fresh paired seeds compare uniform sampling, the common-scaffold $p(1 - p) = u_2$ score, and u_{16} under identical discounted Beta tracking. The internally frozen primary is target-uniform normalized transition-AUC — evaluation success averaged uniformly over the eight predicates, integrated over the paid-transition axis and normalized by the budget — for $u_{16} - p(1 - p)$, requiring a positive exact two-sided sign-flip test at .05 and a point estimate of at least +.01.

Arm means are .64523 (uniform), .63907 ($p(1 - p)$), .68711 (u_{16}). The primary clears both frozen criteria with 15/20 positive pairs, supporting the deployed- N shape against its $N=2$

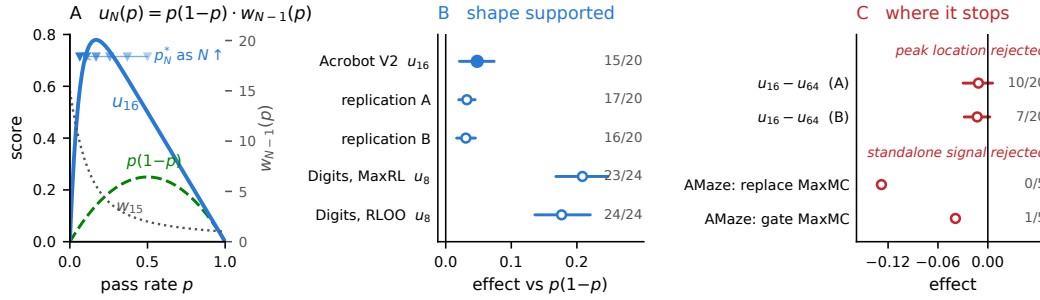


Figure 2: One identity and one claim boundary. (A) Activity is learnability reweighted by MaxRL’s own objective weight (Cor. 1); the peak p_N^* is derived, not tuned. (B) Every measured contrast of the deployed- N shape against its $N=2$ slice, with paired-bootstrap 95% intervals and positive-pair counts; filled marks the one frozen confirmatory primary (§4.1). (C) The two preregistered boundaries. AMaze entries are five-seed development means at one-sixth budget and carry no interval; a full-budget confirmatory campaign is preregistered separately.

Table 2: Fresh Acrobot tournament. Paired differences use 20 fresh seeds under the source/runtime-locked protocol. The two uniform comparisons form one Holm family.

contrast	mean Δ	95% boot. CI	exact p	Holm p	decision
$u_{16} - p(1-p)$	+0.0480	[+.0209,+.0738]	.0034	—	supported by rule
$p(1-p)$ —uniform	-.0062	[-.0226,+.0122]	.5078	.5078	not supported
u_{16} —uniform	+.0419	[+.0218,+.0606]	.0008	.0016	supported

ablation in this fixed family — while $p(1-p)$ itself is not supported over uniform by the frozen secondary.

4.2 Does the deployed- N peak location matter?

The score’s peak $p_N^* = 1 - N^{-1/(N-1)}$ falls $.500 \rightarrow .169 \rightarrow .064$ for $N = 2, 16, 64$, so in every study above the winning arm is also the harder-peaked arm. “Harder peaks beat $p(1-p)$ ” and “the deployed- N peak is correct” predict identically in all of that data, and only the second would be an advance over the $p(1-p)$ literature. We separate them with two preregistered campaigns that hold the deployed estimator at $N = 16$ in every arm and vary only the sampling exponent, on the same 20 logical seeds and the same frozen scaffold, analyzed once each.

An over-shooting arm scored by u_{64} does not lose to u_{16} : the paired difference $u_{16} - u_{64}$ is $-.0113$ (95% CI $[-.0297, +.0057]$, exact $p = .235$, 10/20) on one platform and $-.0128$ ($[-.0284, +.0024]$, $p = .131$, 7/20) on a second. A seven-point exponent sweep $N_{\text{score}} \in \{2, 4, 8, 16, 32, 64, 128\}$ gives arm means $.644, .668, .665, .675, .679, .688, .681$, with argmax at u_{64} , not at the deployed u_{16} , and Spearman rank correlation $+.93$ between exponent and mean. The curve rises past the deployed N : peak-location specificity is not supported at this operating point.

The same campaigns replicate the shape effect. Against $p(1-p)$, u_{16} wins by $+.0322$ ($[+.0198, +.0449]$, exact $p = .000105$, 17/20) and $+.0307$ ($[+.0166, +.0453]$, $p = .000507$, 16/20), versus $+.0480$ in V2 — the first reproductions outside the machine that produced V2, both with a smaller p -value than V2’s own. Magnitude is platform-sensitive (about 35% attenuation on identical seeds between architectures), so we attribute each number to its platform. Because the two new campaigns share seeds and a deterministic engine, their agreement is a portability check, not additional independent seeds.

Why the curve rises past the deployed N . Branching an exact-gradient pool to measure true continuation utility $U_H(x) = J(\text{Train}_H(\theta; x)) - J(\theta)$ at $H = 8$, u_{16} ranks it far better

than $p(1-p)$ (+.106, 10 seeds, exact $p = .002$), while under RLOO — whose mass is $2p(1-p)$ — the two are indistinguishable (+.0001). But U_H peaks near $p \approx .02$, below both p_{16}^* and p_{64}^* : activity ranks utility well and locates it poorly. That local activity and downstream benefit can diverge is also reported across domains under GRPO (Yang et al., 2026); we observe it within one family at a fixed deployed N . Mechanism on 36 synthetic tasks, not evidence for a sampler.

The contrast stays descriptively positive on sampled-group (+.05294) and optimizer-update (+.05366) axes, and u_{16} receives 2.40 fewer groups and 3.42 fewer updates per million transitions, so it did not win by receiving more updates; these non-confirmatory axes identify no causal mechanism. Native-success transition-AUC is .31085, .30521, .37615 across arms. App. C reports the near-equal observed mass means and the audited but RNG-confounded precursor.

4.3 Where the score is starved: activity as a PLR priority on AMaze

The natural competitive test drops the activity priority into minimax’s robust PLR (Jiang et al., 2021; 2023) in place of MaxMC, running the authors’ shipped maze configurations verbatim. Our development sweep — seeds disjoint from any confirmatory block, 5,000 of the 30,000 shipped updates, five paired seeds per arm, held-out return over the three shipped test mazes — is negative, and the reason is informative. Upstream robust PLR scores .629 and the no-curriculum control .539; every activity-priority variant is below the control. Replacing MaxMC with $\mathbb{E}[u_N(p) \mid \mathcal{D}]$ at the shipped 32×1 batch scores .500 ($N=8$; 0/5 seeds beat MaxMC, exact $p=.0625$) and .551 ($N=16$); a 4×8 regrouping that buys per-level evaluation fidelity with level diversity costs $-.207$ for MaxMC too, and the same holds inside ACCEL.

The cause is information, not shape. MaxMC reads the critic’s disagreement $\max\text{-return} - V(s_t)$ at every timestep, continuously, with memory of the best return ever achieved on that level. At the shipped $n_eval=1$ a success-rate score receives one terminal Bernoulli per level visit — a two-valued posterior, nearly all ties under rank prioritisation — and $\max_episode_steps=250$ against a 256-step rollout forbids a second episode; accumulating over revisits does not close the gap. Gating MaxMC by activity instead of replacing it recovers most of the gap (.590; +.089 over replacement, 4/5 seeds; $-.039$ to upstream, 1/5) but does not clear it: u_N serves as a gate on a richer signal, not as a signal of its own. The AMaze student is PPO with GAE and never forms binary rollout groups; the faithful mapping, giving it the MaxRL group estimator, is outside this evidence.

4.4 Externally recorded fresh-wave maze result

We retain the maze factorial as a secondary estimator measurement; it does not test the acquisition score. A 1.26M-parameter transformer is trained on procedurally generated 17×17 mazes in balanced blocks that share seed, warm start, task pool, and evaluation schedule across $\{\text{MaxRL}, \text{GRPO}\} \times \{\text{uniform}, \text{FrontierMax}\}$, at $N = 32$, eight prompt groups per step, 250 steps, evaluation every 25 steps on 16 held-out mazes per level, with identical deployed optimizer settings and no per-estimator learning-rate tuning. The outcome is time-integrated pass@8 coverage relative to the warm start. The first factorial rejected its stronger externally recorded endpoint claim (3/6 and 4/6 blocks with the predicted sign) and we retracted the earlier zero-exception cohort claim; wave-1 time-integrated coverage was positive in all blocks but discovered exploratorily, so the external record specifies it as wave 2’s primary with a falsification rule before new seeds ran, the locking commit object being unvended here.

The wave-2 primary is positive in all six fresh blocks under each sampler (exact two-sided sign $p = .03125$ per sampler; 6/6 is the only outcome clearing .05 at six blocks). The inferential anchor is the block-averaged mean +.0195 (post-hoc 95% t interval $[+.0115, +.0275]$; every leave-one-block-out interval above zero); all 12 cross-wave block averages are positive, descriptively so because wave 1 selected the outcome. This supports an estimator-conditioned coverage ordering at common optimizer settings, not universal estimator superiority. Easy-band localization is suggestive only (four positive blocks, one tie, one negative; interval

crossing zero), and the frozen AUC robustness multiverse awaits checkpoint trajectories that remain external (Fig. 3, App. C).

Secondary records. Paid-probe selection, failure recycling, and the historical Countdown aggregate are reported only in App. C and support no named-method superiority claim; their measurement lesson is that raw per-task outcomes are needed to distinguish mean accuracy from standard pass@ k .

5 Related work

Objective-side work such as MaxRL (Tajwar et al., 2026) and RL2ML (Zheng, 2026) characterizes which population or finite-rollout objective a group estimator implements, typically holding the task distribution fixed. Learnability curricula such as ProCuRL (Tzannetos et al., 2023), SFL (Rutherford et al., 2024) and LILO (Foster et al., 2025) adapt that distribution using success variability, while advantage-bandit curricula such as SEC (Chen et al., 2025) and DUMP (Wang et al., 2025) estimate activity empirically from observed policy advantages, and TAC (Yang et al., 2026) augments local activity with first-order cross-domain transferability under GRPO. We study a complementary object: the exact task-level finite-group activity geometry a deployed estimator induces (Def. 1), derived from its algebra before any curriculum hypothesis is tested, with a peak that migrates with the rollout budget and used for task selection rather than objective redesign. Superiority over the methods below is untested and not claimed.

Concurrent exact analyses of group estimators. A concurrent cluster derives closed-form expected update magnitudes on the standard-deviation-normalized branch: the group-SD identity with DAPO’s silent-group mass $p^G + (1 - p)^G$ (Bay, 2026), an expected absolute GRPO advantage of $2\sqrt{p(1 - p)}$ motivating 50%-success targets inside a learned-curator bandit (Gu et al., 2026), and energy- or variance-based selection scores making the same claim empirically (Cui, 2026). There the per-rollout magnitude peaks at $p = .5$ independent of N , which enters only through the silent-group factor; the practical-MaxRL mass is instead the estimator’s own unnormalized pass@ N gap, with a peak migrating as $\ln N/N$ — the functional form our sweeps test. MoPPS (Qu et al., 2026) anticipates the posterior machinery (streaming Beta posteriors, Thompson-sampled pass rates, targeting $p \approx .5$); our novelty claim is only the utility sampled toward.

Curricula and rollout allocation. Intermediate-difficulty sampling predates this work: under ProCuRL’s experimental assumption that optimal-policy success is one, its proximal score reduces to $p(1 - p)$ (Tzannetos et al., 2023), and SFL (Rutherford et al., 2024) and LILO (Foster et al., 2025) prioritize binary reward variance. Our distinction is convention-specific: $u_N = A_N/2$ is the arbitrary- N expected half-mass of the deployed centered MaxRL rule, derived from its algebra rather than estimated online. VCRL (Jiang et al., 2025) thresholds realized group variance; PLR (Jiang et al., 2021) and ALP-GMM (Portelas et al., 2020) use TD error, staleness, or temporal learning progress, and so react to forgetting where instantaneous coefficient mass cannot — our AMaze arm is robust PLR (Jiang et al., 2023) with the replay priority replaced by the posterior activity and nothing else changed (§4.3). Generator-based UED — PAIRED (Dennis et al., 2020), ACCEL (Parker-Holder et al., 2022) — creates tasks rather than pricing a fixed pool. GRESO (Zheng et al., 2025) and DPS (Mao et al., 2026) select prompts before rollout from historical or predicted reward dynamics, VIP (Nguyen et al., 2026) varies per-prompt rollout counts to cut predicted gradient variance, and adaptive allocators stage extra rollouts toward high-value prompts (HORA (Wang et al., 2026), VIGOR (Jiang et al., 2026), DARS (Yang et al., 2025), Reinforce-Ada (Xiong et al., 2025)); FrontierMax instead selects the task distribution before generation at fixed N . BBG (Yuan et al., 2026) gates updates by posterior marginal pass@ k contribution and boundary-aware Curriculum RL (Cai et al., 2026) adds external guidance near an estimated reasoning boundary. These target hit probability, variance, exploration, boundary metrics, or signal creation — not practical-MaxRL coefficient mass. RL2ML (Zheng, 2026) separates finite-rollout objectives, update scale, variance, and metric gain; its $T = N$ alignment retains $K = 0$ via the direct estimator or a score-baseline control variate, while our

$T = N - 1$ statement is for the deployed hybrid that centers only when $K > 0$ and zeros $K = 0$.

Coverage measurement. RLVR can improve pass@1 while reducing high- k coverage (Yue et al., 2025; Yuan et al., 2026; Zhou et al., 2026); entropy collapse is complementary evidence (Cui et al., 2025) and pass@ k objectives address the metric directly (Chen et al., 2025). We ask whether an external curriculum changes that coverage under two estimators. The secondary hindsight and relabeling record (App. C) is not evidence for the acquisition score, but it marks the limit of allocation: a sampler redistributes existing signal, while relabeling or subproblem decomposition (Jiang et al., 2026) can create signal outside its support — at $A_N = 0$ no priority helps.

6 Limitations

Coefficient mass ignores score-gradient direction, parameter sharing, optimizer state, task transfer, and non-i.i.d. rollouts: it is a sampler-facing surrogate, not a guarantee of loss reduction or metric improvement. The fixed-completion N -sweep is post-guidance (designed after the theoretical guidance existed, hence not preregistered) on one co-designed synthetic chain, and the exact-probability Digits counter-test is a contextual bandit rather than trajectory RL. The fresh V2 Acrobot score test uses one fixed eight-threshold family, one 640-parameter learner, and no held-out task distribution or native named-method implementation; its 20 seeds were inherited from a historical precursor rather than set by prospective power analysis, and its exact paired test requires sign exchangeability under the sharp null. V2 paired no mismatched- N score against a fixed deployed N ; §4.2 does, and finds no peak-location specificity, so the evidence supports rollout-aware difficulty targeting and specifically not the deployed- N peak. The four evidence families froze their primaries independently at $\alpha = .05$; we claim no paper-level error rate. Retained evaluation records do not include per-episode trajectories. The maze result uses a small transformer, one task family, and common rather than estimator-tuned learning rates; its easy-band mechanism remains descriptive. The Coefficient-Activity PLR row in Table 1 is a development negative (§4.3), not preregistered evidence, and we claim no advantage over PLR, robust PLR, AC-CEL, or any other minimax baseline. No robotics performance is reported; the paid-probe, recycling, Countdown, gate, and transfer records are secondary and confined to App. C.

Registration timing is the recurring gap. The Digits lock and the Acrobot tournament lock are internally hashed — the latter binds source, runtime, gates, and artifacts — but neither is backed by an immutable public pre-execution commit independently establishing timing, and the external maze wave-2 record, though specified before execution, has no locking commit object vendored here: the release can audit stored protocols and verdicts, not their timing. Artifact coverage is likewise incomplete, with maze checkpoint trajectories external and the Digits replay payload local without a download URI; App. D itemizes these gaps alongside the 53-row compact registry and the distinct release-branch audit object. Together they limit auditing and preclude a broad neural claim.

7 Conclusion

The estimator belongs in the evaluation of data-selection design: its coefficient algebra fixes which tasks can emit activity at a chosen rollout count, and a separately specified coverage ordering held under both tested samplers. In a fixed classical-control family u_{16} beat both u_2 at the same deployed $N=16$ and uniform sampling, replicating on two platforms, while Digits shows the estimator-to-sampler mapping is not universal. Two preregistered follow-ups drew the boundary: with the estimator held at $N=16$, performance rose past $N=16$, so the harder-peaked shape helps and the deployed- N peak location does not; and as a standalone PLR priority on AMaze, activity is starved at one Bernoulli per level visit of the per-timestep signal a regret proxy reads for free. Each group estimator induces its own activity geometry, but activity diagnoses available update, not learning utility: evaluate data selection with the estimator it ships on, report pass@ k from retained outcomes, and treat coefficient mass as a gate where activity binds, not a universal law.

Reproducibility statement

The repository includes exact-enumeration tests, a one-command artifact check, structured seed-block analyses, stored protocol and verdict documents, figure derivation scripts, a source-locked 60-run Acrobot tournament with deterministic retained-ledger reanalysis, and a generated 53-row compact registry for locally indexed maze, Countdown, and GSM8K artifacts. Secondary records and receipt manifests remain documented in the appendix. Registration timing objects that remain external are disclosed rather than treated as locally auditable. The independent unit is stated for every central result; known raw-data gaps and secondary-study provenance are disclosed in App. D.

AI use statement

AI assistants supported planning, implementation, debugging, literature discovery, statistical and artifact auditing, figure preparation, and manuscript editing. The authors selected the questions and decision rules, ran the studies, verified derivations, code, numerical results, and citations, and take responsibility for every claim.

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A Proof details

Truncation order. For $K > 0$, conditioning on one sampled response and using exchangeability gives

$$\mathbb{E}\hat{g} = \left(\frac{1 - (1 - p)^N}{p} - (1 - p)^{N-1} \right) \nabla p.$$

Since $1 - (1 - p)^N = p \sum_{j=0}^{N-1} (1 - p)^j$, subtracting the final term leaves $\sum_{j=0}^{N-2} (1 - p)^j \nabla p = \nabla J_{N-1}$. If the centered control-variate term is retained on $K = 0$, its contribution is $(1 - p)^{N-1} \nabla p$. The expectation becomes $\sum_{j=0}^{N-1} (1 - p)^j \nabla p = \nabla J_N$. Thus the direct/non-dropped convention has order N , while the deployed hybrid convention above has order $N - 1$; theorem and implementation must use the same convention. At audited MaxRL source commit 7197bbb, the released path computes $(r_i - \bar{r})/(\bar{r} + \epsilon)$ and zeros all-failure groups; after the deployed gradient average, its $\epsilon \rightarrow 0$ coefficients match this convention up to common scale. The parsed truncation-order setting is inactive on that path.

Coefficient mass. For $0 < K < N$, each success has coefficient $(N - K)/(KN)$ and each failure has coefficient $-1/N$. The two classes each contribute $(N - K)/N$ in absolute value. For $K \in \{0, N\}$ the deployed centered estimator contributes zero. With $K \sim \text{Bin}(N, p)$,

$$\begin{aligned} A_N(p) &= \frac{2}{N} \sum_{k=1}^{N-1} (N - k) \binom{N}{k} p^k (1 - p)^{N-k} \\ &= \frac{2}{N} \{ \mathbb{E}[N - K] - N \Pr[K = 0] \} \\ &= 2\{1 - p - (1 - p)^N\}. \end{aligned}$$

The peak follows from $u'_N(p) = N(1 - p)^{N-1} - 1$ and strict concavity for $N > 1$.

Posterior-integrated activity. For $p \sim \text{Beta}(a, b)$, $\mathbb{E}[p] = a/(a + b)$ and the beta-function ratio gives

$$\mathbb{E}[(1 - p)^N] = \frac{B(a, b + N)}{B(a, b)} = \frac{(b)_N}{(a + b)_N}.$$

Substitution in $u_N(p) = 1 - (1 - p)^N - p$ proves the deterministic posterior priority in §3. Subtracting it from $u_N(\mathbb{E}[p])$ gives the printed nonnegative Jensen gap.

Estimator limits. Under the deployed $1/N$ gradient-average normalization, RLOO expected half-mass is $p(1 - p) = u_2(p)$, so $u_N(p)/u_2(p) \rightarrow N - 1$ as $p \rightarrow 0$. Near mastery, both are first order in $q = 1 - p$. With sample-SD normalization, GRPO group mass is proportional to $\mathbb{E}[\sqrt{K(N - K)}]$ up to deployed normalization constants; its relative tail scaling is verified by exact enumeration in `verify_guidance_math.json`. These are coefficient statements, not gradient-direction statements.

Dependent rollouts. The per-group identity does not require independence. For any binary group,

$$\mathbb{E} \sum_i |w_i| = 2\{\Pr(K \geq 1) - \mathbb{E}[K]/N\}.$$

With a common marginal p , the second term is p ; independence is used only to replace $\Pr(K \geq 1)$ by $1 - (1 - p)^N$. In a post-guidance exact beta-binomial CPU stress test, count-distribution enumeration matched the general identity to 2.58×10^{-13} . At $\rho = .10$, the activity peak moved from .169 to .234 for $N = 16$ and from .106 to .186 for $N = 32$. This dependence model shows how increasing positive correlation can lower activity inside that family; pairwise correlation alone does not determine a general joint law. The result is a scope diagnostic, not evidence that coefficient mass causes learning or that beta-binomial dependence describes neural rollouts.

Posterior-mean bias and the uncertainty penalty. With $p_x \mid \mathcal{D} \sim \text{Beta}(a_x, b_x)$, strict concavity of u_N for $N > 1$ and Jensen’s inequality give $\mathbb{E}[u_N(p_x) \mid \mathcal{D}] \leq u_N(\mathbb{E}[p_x \mid \mathcal{D}])$, so a priority computed from the posterior mean overstates expected activity. The gap is nonnegative and closed-form,

$$u_N\left(\frac{a_x}{a_x + b_x}\right) - \mathbb{E}[u_N(p_x) \mid \mathcal{D}] = \frac{(b_x)_N}{(a_x + b_x)_N} - \left(\frac{b_x}{a_x + b_x}\right)^N,$$

and vanishes as the posterior concentrates. It is an uncertainty penalty, not an exploration bonus: it shrinks the priority of tasks whose pass rate is still poorly determined.

B Algorithms and estimator contracts

Posterior teacher. Each task stores discounted successful and failed rollout counts. A Thompson draw $\tilde{p}_x$ is mapped through u_N and mixed with a uniform floor. Teacher state, sampler cursor, and checkpoints restore together. Observations use pre-relabel rewards so recycled outcomes do not inflate the requested task’s posterior.

Coefficient-Activity PLR (development negative). The AMaze implementation is a narrow overlay on robust PLR in the pinned minimax substrate (Jiang et al., 2021; 2023). It uses cumulative Beta evidence per replay-buffer slot and replaces MaxMC priority with deterministic $\mathbb{E}[u_N(p) \mid \mathcal{D}]$; the declared N may equal `n_eval` (exact grouping) or be decoupled from it. Buffer admission, the robust replay gate, inverse-rank mass, staleness mixing, and with-replacement draws remain PLR mechanisms. Tie-aware ranking is shared with the matched MaxMC control. §4.3 reports the development sweep: the replacement priority does not beat upstream robust PLR, and no baseline-performance claim is made.

Relabeling (secondary). For an all-fail request, a verified achieved goal can create an auxiliary update where repeating the request would emit none. A parser proposes that goal from a failed response, and the environment’s exact verifier must accept the response under that goal. Prompt conditioning and explicit response goal declarations are rewritten; answer spans, arithmetic, decimals, signed values, and unknown contexts are preserved. The reported-run implementation is frozen, while a conservative release wrapper and 26 CPU regression/property tests protect future runs. Relabeled groups remain auxiliary selected-data updates; no importance ratio converts them into fresh on-policy target samples. The intervention changes both update dose and direction, and its achieved-goal distribution is biased toward outputs the policy already reaches (Andrychowicz et al., 2017; Ghosh et al., 2021).

C Experimental protocols and secondary results

Fixed-completion N -sweep. The environment contains three nested 12-level chains and ten actions. Every cell runs exactly 51,200 completions, with arms for uniform, $p(1 - p)$, and u_N sampling under the practical MaxRL estimator, no hindsight, and eight paired seeds. Checkpoints share a completion clock. The post-guidance analysis is descriptive.

Digits factorial (internally frozen negative). On a stored 1,077/360/360 train/development/test split of sklearn Digits, a 4,810-parameter float64 policy recomputes each example’s exact correct-class probability before every update. We crossed practical MaxRL and RLOO with uniform, $p(1 - p)$, and u_8 sampling. Five-rate development selected learning rate .1 for MaxRL, RLOO, and the common-rate check. Across 24 fresh paired blocks, the internally frozen interaction was +.01589 (paired bootstrap 95% CI $[-.01686, +.04712]$, exact two-sided sign-flip $p = .350$; 15/24 positive), so it was not supported despite passing the treatment-delivery gate. MaxRL did favor u_8 over $p(1 - p)$ by +.20842 (CI $[+.16791, +.24744]$; 23/24), but RLOO’s frozen prediction reversed: $p(1 - p) - u_8 = -.17665$ (CI $[-.22042, -.13593]$; 0/24). Moreover, the matched samplers were below uniform for both MaxRL ($-.11279$) and RLOO ($-.37581$). Because every selected rate equaled .1, the tuned and common schedules have byte-identical ledgers and

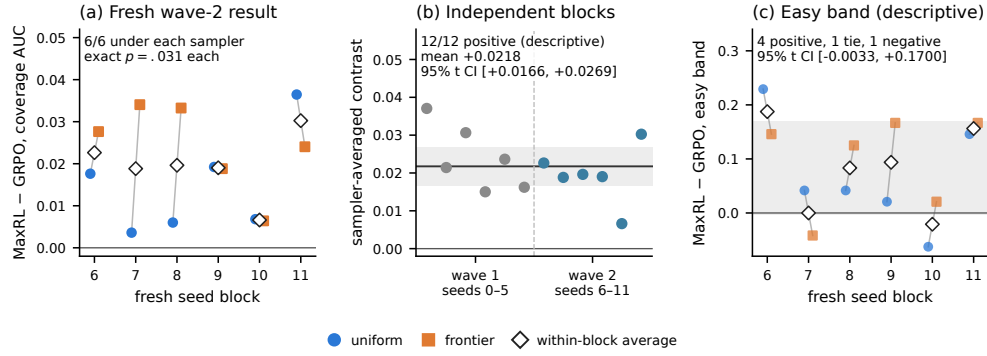


Figure 3: Maze factorial at the independent seed-block unit. (a) The externally recorded fresh-wave result is positive in 6/6 blocks under each sampler (exact sign $p = .03125$ each). Connected observations share a block. (b) Sampler-averaging gives wave-2 mean $+0.0195$ and post-hoc 95% CI $[+.0115, +.0275]$; 12/12 cross-wave block averages are descriptive. (c) Easy-band localization is suggestive: four positive blocks, one tie, one negative, and an interval containing zero.

checkpoints and identical scientific summary content after removing phase/authorization labels; they are identity checks, not independent replications. This exact- p contextual bandit therefore supports neither the planned interaction nor a universal curriculum advantage; it shows why expected coefficient mass must remain an estimator-activity proxy rather than a learning law. Its internally hashed local lock is not backed by an immutable public pre-execution commit that independently establishes timing.

Acrobot V3 (historical). The frozen control uses official Acrobot-v1, eight nested thresholds from -1.5 through native success at 1.0 , one shared H64 actor, practical MaxRL, $N = 16$, and no hindsight. Uniform and u_{16} receive 20 paired seeds and a nominal two million actual transitions per run, completing the final group. Evaluation uses 32 shared trajectories total, reused across all eight predicates, at initialization and each 100,000 transitions with fixed common random numbers. The internally frozen primary is target-uniform normalized transition-AUC; group diagnostics and evaluation curves for all 40 runs are vendored. A post-hoc audit found cross-seed overlap between actor-initialization and action-sampling RNG roots, so the retained effect and mechanism audit are descriptive rather than clean confirmation. The mechanism audit is deterministic from those records.

Acrobot fixed-pool tournament V2. The replacement reruns uniform and u_{16} and adds $p(1 - p) = u_2$ on fresh logical seeds 20000–20019. All 60 source/runtime-locked runs completed after an outcome-blind development gate; globally disjoint RNG roots and per-run group-diagnostic and aggregate-checkpoint ledgers were retained. The primary $u_{16} - p(1 - p)$ transition-AUC difference is $+0.04803$ (paired bootstrap 95% CI $[+.02094, +.07385]$, exact sign-flip $p = .003361$), clearing the predeclared $+0.01$ estimate filter; Table 2 gives the Holm-controlled uniform comparisons. Observed mass means are close between adaptive arms ($.7344$ versus $.7490$ per group; 115.57 versus 115.90 per million transitions), both descriptive paired intervals include zero, and no equivalence test was frozen. The p -value tests zero rather than the SESOI, and pairing does not guarantee sign exchangeability. The $p(1 - p)$ arm isolates only the score on our scaffold, not full ProCuRL or SFL.

Paid-probe ProCuRL selection semantics. This is a separate 80-pair family, not an extension of the V2 table. It keeps the same eight thresholds, H64 actor, practical MaxRL estimator, $N = 16$, and 3×10^{-4} learning rate. Its four arms are ordinary uniform; a uniform sham with paid probes; source-faithful ProCuRL selection $\text{softmax}\{20\hat{p}(1 - \hat{p})\}$; and continuous-range-matched u_{16} selection. Probed arms run 20 episodes per task initially and at every crossed 5,120-student-transition boundary; all student and probe transitions count toward the nominal two-million paid budget. Across seeds 21000–21079, the

internally frozen fixed-paid-AUC contrast u_{16} -ProCuRL is $+0.004894$ (paired SD $.022139$, $t_{79} = 1.9773$, $p = .05149$; 20,000-resample CI $[+.000110, +.009727]$; one-million sign-flip $p = .05161$). It misses both the .02 SESOI and the .05 paired- t criterion. The five Holm-family contrasts are ProCuRL-sham $-.001704$ (adjusted $p = .4568$), u_{16} -sham $+0.003190$ ($.4547$), ProCuRL-ordinary $-.313774$ (3.47×10^{-66}), u_{16} -ordinary $-.308880$ (9.48×10^{-67}), and sham-ordinary $-.312070$ (4.06×10^{-68}). Mean fixed-paid AUCs are $.33771$ (ProCuRL), $.33942$ (sham), $.65149$ (ordinary), and $.34261$ (u_{16}).

The probed arms average 28.0–28.1 sweeps and 140.5–141.1k student transitions versus 2.003M for ordinary uniform; 93.19–93.21% of their paid transitions are probes. They average 15.4–19.8 optimizer updates versus 259.1 ordinary-uniform updates, with mean paid-budget overshoot 69,960–72,939 transitions versus 2,984. All 320 confirmation and 12 outcome-blind development runs reconcile exactly, all 11 development gates pass, and the common-random-number invariants replay.

An earlier outcome-blind development wave was invalidated before any gate or arm contrast because 73 entropy-sum replays differed by at most 7.28×10^{-12} . Its seeds were burned and its incident receipt is retained; the reported family uses fresh seeds and a replacement lock.

The external confirmatory raw is 1,374,886,097 bytes. Its SHA-256 boundary is:

```
raw: b1f8756c249effab8c77101c8bca73ddf708a5e143c18fe8742fd5712fdd7c12
lock: b7c7f76f6aaffa1fe65557717bfe545f2ec850495d370cd97fe72a8871fc8d0f
analysis: 2010e30b5b15a212e2d6bdfaacd43d2434e5f468a96be613cf59744b9bc2fb38
```

The source bundle carries a 320-record external-raw manifest: compact mode checks its small-artifact bindings, while full verification requires the raw file and replays locked validation. The local lock is mechanical provenance, not an immutable public pre-execution registration.

Post-guidance rollout-allocation factorial. We also compared fixed $N = 16$, published-HORA-style posterior hit allocation (Wang et al., 2026), and a mass-aware marginal that counts four already-spent probe rollouts. A 2-sampler by 3-allocator CPU factorial uses 16 paired seeds and 51,200 completions per cell. The mass-aware arm improves pass@8 AUC over HORA-style allocation by $+0.00676$ (11/16 positive pairs) and over fixed groups by $+0.02517$ (14/16). However, it reduces realized coefficient mass per completion by $.000306$ relative to HORA-style allocation in all 16 pairs, falsifying the frozen mechanism prediction. Its group sizes are also extreme (cell-average maximum about 76 versus 57 for HORA and 16 fixed). We therefore treat the result as exploratory motivation for posterior-calibration and allocation-cap ablations, not a fourth contribution.

A frozen follow-up crossed two samplers, the two adaptive scores, four caps, and three information sources over 16 paired seeds (50 cells; 800 runs). Restricting to the 32 deployable-information adaptive cells, all 32 adaptive-minus-fixed mean pass@8-AUC contrasts were positive, with 22 unadjusted descriptive Student- t intervals above zero; no multiplicity-adjusted inference is assigned. Yet realized coefficient mass was lower in 28/32 cells, with 24 intervals below zero and none above zero. The remaining 16 oracle-preupdate cells are diagnostic. Thus the expanded result reinforces the learning-mass mismatch rather than a mediation claim. The frozen engineering rule chose cap 32: averaged over samplers for the fresh-group/same-step cells, it reduced maximum group size by 58.07% with an AUC change of $-.00271$ versus uncapped. This is a point-estimate filter for a future pilot, not an equivalence test. The source/runtime lock is internally hashed but lacks a public pre-execution commit establishing timing.

Maze factorials. Wave 1 has six shared seed/warm-start blocks and six arms per block: MaxRL and GRPO under uniform and FrontierMax sampling plus two diagnostic GRPO arms. Its recorded endpoint hypothesis failed. Wave 2 has six fresh blocks and the four core arms. The external execution record names time-integrated pass@8 relative to warm start primary, with at least 5/6 positive blocks required under each sampler. The block analyzer reports sampler-specific exact sign tests, sampler-averaged contrasts, post-hoc intervals, and leave-one-block-out checks. No pooled p -value treats sampler observations as independent.

AUC robustness status. The frozen multiverse covers integration rule, warmup inclusion, horizons, and leave-one-checkpoint-out variants, but all 24 wave-2 checkpoint JSONLs and warm starts remain external. Structured summaries recover the anchors (uniform 6/6, mean $+0.01496$; FrontierMax 6/6, $+0.02404$; sampler-averaged 6/6, $+0.01950$), but cannot identify those variants; the analyzer therefore exits explicitly blocked.

Countdown. We fine-tune SmolLM2-360M with $N = 16$. The v2 pool uses operand permutation, parenthesization, and exact division. Tier 1 has 128 evaluation tasks and 16 verifier-scored samples per task. B1 disables recycling; B2 admits exact-verifier achieved goals; the committed scoreboard contains three-seed aggregates. ARM B disables recycling and uses two optimizer epochs on every live group; its three individual endpoints are vendored. Headline endpoints are step-60 mean@16 and VERL bootstrap best@16. The latter uses 1,000 with-replacement resamples of size 16 per prompt and is not standard unbiased pass@16; missing per-task outcomes prevent recomputation. A stored task-identity summary reports zero tier-1 and tier-2 SFT overlap and 27 exposed tier-0 tasks; the source manifests needed to recompute those counts are absent. Tier 0 is excluded from absolute claims.

The reported recycling-package aggregate moves tier-1 mean@16 from $.278 \pm .066$ to $.324 \pm .014$ while bootstrap best@16 moves from $.541 \pm .024$ to $.492 \pm .013$ (mean $\pm$ sample SD across three seeds). Complete three-seed B1/B2 records are not vendored, so this supports neither a paired seed test nor a named-method effect claim. It is retained only as a measurement caution: mean accuracy alone cannot reveal whether coverage was spent to buy it.

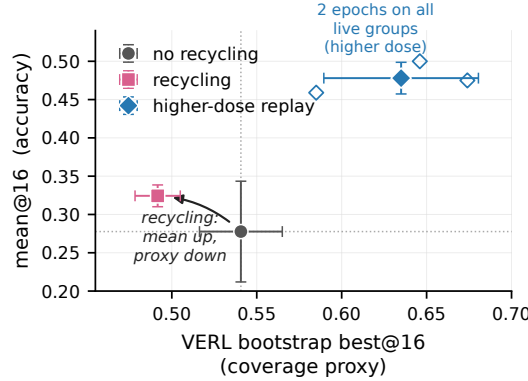


Figure 4: Secondary Countdown accuracy-coverage-proxy record. Error bars are sample standard deviations across three seeds; seed 3 for B1/B2 is aggregate-only. The recycling-package aggregate has higher mean@16 and lower VERL bootstrap best@16. The live-group replay control uses two optimizer epochs on every live group, so it cannot separate dose from update direction.

The externally specified replay control reaches mean@16 $.478 \pm .021$ and bootstrap best@16 $.635 \pm .046$, with all three endpoints vendored. Recycling permits at most 12 auxiliary groups per 64 requested groups (18.75%; realized dose can be lower), whereas replay doubles optimizer epochs on every live group. This higher-dose alternative improves both logged metrics without identifying a dose-monotonic causal bound.

Negative branches. The artifact retains the failed maze endpoint, gate, Jugs, and robotics branches. GSM8K missed its delivery threshold ($.601480$ versus $< .60$) and is inconclusive; the single-seed Countdown per- k curve is exploratory. The corrected-code gate dose-response (frozen protocol, three paired seeds, independent A100 environment, retained 16-bit raw outcomes per task) passed its transfer gate — ungated recycling beat no-recycling on mean@16 in 3/3 seeds — and its manipulation check (rejection graded 0/.72-.80/.88-.90/.93 across gate strengths), but neither tested setting met the frozen usefulness criterion; per-

seed signs were mixed at $n = 3$. Its raw outcomes also record, descriptively, one seed in which mean@16 rose while standard observed-set pass@16 fell $.656 \rightarrow .414$ — a non-proxy instance of the ambiguity the operational recommendation targets.

D Artifact and data accounting

bash reproduce.sh -build tests the identities and integrations, verifies declared manifest inputs and the local compact registry, runs stored endpoint spot checks, regenerates figures, and builds the canonical main_iclr2027.pdf plus the extended main.pdf. The compact registry has 53 records: 35 maze, 11 Countdown, and 7 GSM8K. The broader 562-row registry used during release-branch auditing is a different object; it is not copied over this checkout’s registry and is not used for the local count.

Missing external data are maze checkpoints; Countdown task manifests, outcomes, and B1/B2 seed 3, so the clean 101-task tier-0 analysis cannot be reconstructed; the 1.375 GB paid-probe confirmatory raw (its content hash and per-run receipt manifest ship); and the Digits 5.08 GB replay payload (block contrasts and a content manifest ship). Compact inputs and checksums are in paper/results/manifest.json and fail closed.